

FloodMesh field test

Team guide · read your own page · ask the leader if anything is unclear

Device	Your job	Where	Name
A	Person in trouble. Sends the SOS.	Stays at the spot the leader gives	
B	Walker. Walks and checks the signal.	Moves around the area	
C	Roof tower. Passes messages on.	On the building roof. Never moves	
E	Rescuer. Gets the SOS and replies.	Moves around, finds A	

D stays at home. Keep it switched off.

Before you start

1. Antenna **screwed on tight**. Never switch the device on without its antenna.
2. Switch it on. The top of the screen shows your letter, for example **A USER** or **A CIV**.
3. Battery: the home screen shows a voltage like **3.85V**. Below **3.5V**, tell the leader.
4. On your phone open **floodmesh-field.vercel.app**, pick your letter, type the passcode the leader gives you. Tap **Share my location** and allow it. Keep that page open, screen on.
5. Join the team **WhatsApp group** and keep the phone charged.

Golden rules

- **Do not hold any key while switching on.** Some keys start special modes; ***** and **0** held down wipes the device.
- **Do not press * and # together** unless your page says so. That sends an SOS.
- Screen went dark? Press **D**. The first press only wakes it.
- Keep the antenna **pointing up**. Do not cover it with your hand or keep it in a pocket.
- The red box on the website saying **“Nothing received yet”** is normal today. Ignore it.

The three numbers you will read

Number	What it means	Good	Weak	About to fail
dBm	How strong the signal is. Closer to 0 is stronger.	−60 to −90	−100 to −110	−115 to −125
SNR	How clear the signal is above the noise.	above 0	0 to −5	−7 or lower
hop	How it reached you.	0 = direct	1 = through one device	2 = through two

The keypad

1	2	3	A SOS list (E only)
4	5	6	B Status
7	8	9	C Inbox · OK
*	0	#	D back · wake

To do this	Press
See the signal list (Status)	Home → B
Read messages (Inbox)	Home → C , pick one with A / B, open with C
Send an SOS	Hold * and # together for 3 s. Then press 1 Medical, 2 Evacuation, 3 Hazard, 4 Food, 0 General. Press nothing for 5 s = General. D = cancel
Stop your own SOS	Hold D on the home screen
Write a message	Home → 0 , type, hold C to send. # switches words / letters / numbers
Turn the ping on or off (sends a signal every 30 s)	Status (B) → C
Change power mode (roof device only)	Status (B) → #
Go back / wake the screen	D

Reading the Status screen

Each line is one device your device can hear:

HEARD hop dBm SNR age n

A 0 -98 4.5 12s 7

= A heard **directly** (hop 0), **-98 dBm**, SNR **4.5**, last heard **12 seconds** ago, **7** times so far. If **age** keeps growing past 1 minute, that device is out of range.

Device A: person in trouble

You stay where the leader tells you and send the SOS. Everyone else measures you.

Do this

1. Switch on. Press **B** (Status), then **C**. The top shows **PING**. Your device now sends a signal every 30 s so others can find you. Press **D** to go back.
2. Go to your spot. Stand still, device at chest height, antenna up.
3. When the leader says "**SOS now**": hold * **and** # together until the channel screen opens, then press **1** (Medical). Write down the time.
4. Watch your screen:
 - **DELIVERED** = the rescuer's device got it.
 - **HELP IS COMING** = the rescuer replied.
 - **SOS NOT DELIVERED** = nobody got it. Tell the leader.
5. If the leader asks you to stop the SOS: hold **D** on the home screen.
6. When the leader says "move", go to the next spot and repeat.

Do not

- Do not turn the ping off.
- Do not walk away from your spot without telling the leader.

Send on WhatsApp (and type the same in the website Notes)

A | spot 1 | SOS 10:32 | DELIVERED 10:32 | HELP 10:35

A | spot 2 | SOS 11:05 | NOT DELIVERED

Device B: walker

You walk the route and check how well your device hears A, C and E.

Do this

1. Switch on. Press **B** (Status) and leave this screen open while you walk.
2. Walk the route the leader gives you.
3. About every **100 m**, or at every corner: **stop for 1 minute**. Read the lines for **A**, **C** and **E**: hop, dBm, SNR.
4. Take a **photo of the screen** and send it with one line (below).
5. If **A** disappears (its **age** keeps growing past 1 minute), stop. That is the edge. Tell the leader and note the place.

Do not

- Do not press * and # (that sends an SOS).
- Do not keep the device in your pocket while you read it.

Send on WhatsApp (and type the same in the website Notes)

B | near temple | A: hop 0, -97, SNR 3 | C: hop 0, -88, SNR 8

B | bus stop | A GONE (edge) | C: hop 0, -109, SNR -4

Device C: roof tower

C sits on the roof and passes everyone's messages on. Set it up once, then leave it alone.

Set it up

1. Put C at the **highest open spot** on the roof, **antenna pointing up**. Away from metal tanks, dish antennas and walls. Not inside a closed metal or plastic box.
2. Connect it to a **power bank** with the USB cable. It must never run out.
3. Press **B** (Status). Press **#** until the third line says **PWR ON**. The top of the home screen now shows **P** before the battery. This keeps C awake so it passes on everything.
4. If rain is possible, cover the device loosely with a plastic bag, antenna still pointing up.
5. Leave it. Do not press anything else.

Every 30 minutes

1. Press **D** to wake the screen, then **B** for Status.
2. Take a **photo** of the screen. The second line shows **rly** = how many messages C passed on. The list shows everyone C can hear.
3. Check the power bank is still on.

Send on WhatsApp (with the photo)

C roof | 10:30 | rly 14 | hears A -95, B -101, E -99

Device E: rescuer

You get the SOS, reply "Help is coming", and then find A using the signal.

Check first

The top of your screen says **E RSP TEST**. That means rescuer mode. Only your device shows SOS messages; the others pass them on silently.

When an SOS arrives

1. Your device **beeps for about a minute**. Press any key to stop the beeping.
2. Press **A** (SOS list). Pick the SOS with A / B, open it with **C**.
3. Write down: **time, channel** (Medical, ...), **hop, dBm, SNR**.
4. Reply: on the open SOS, **hold 4 and 6 together for 3 s** until the bar is full. A's screen then shows **HELP IS COMING**.

Then find A

1. Press **D** back to home, then **B** (Status). Watch A's line.
2. Walk. If A's **dBm gets closer to 0** (for example $-105 \rightarrow -90$), you are getting closer.
3. Turn slowly on the spot with your body between you and the signal: the direction where the number is **worst** is behind you.

Send on WhatsApp (and type the same in the website Notes)

E | SOS from A 10:32 | Medical | hop 1 | -104 | SNR -2 | replied 10:35
E | found A 10:48 | last reading -71

Notes sheet (if you prefer paper)

Device: _____ Name: _____ Battery start: _____ V end: _____ V

[illegible]